

# Aprendizaje de Máquina

## U3 Otros Métodos de Agrupamiento

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## Section 1

# Density-based algorithms for large data sets

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# Introduction

In this framework, **clusters** are considered as **regions** in the  $l$ -dimensional space that are **dense** (in some sense) in points of  $X$ .

Most of the density-based algorithms do not impose any restrictions on the shape of the resulting clusters. Thus, these algorithms have the ability **to recover arbitrarily shaped clusters**.

In addition, they are able to handle efficiently the **outliers**.

Moreover, the **time complexity** of these algorithms is **lower than**  $O(N^2)$ , which makes them eligible for processing large data sets.

# Introduction

Typical density-based algorithms are:

- The **Density-Based Spatial Clustering of Applications with Noise (DBSCAN)**.
- The **Distribution-Based Clustering of LArge Spatial Databases (DBCLASD)**.
- The **DENsity-based CLUstEring (DENCLUE)**.

Although these algorithms share the same basic philosophy, they differ in the way the **density is quantified**.

## Section 2

# The DBSCAN Algorithm

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# DBSCAN

The **density** as it is considered in **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** around a point  $x$  is estimated as the number of points in  $X$  that **fall inside** a certain region in the  $l$ -dimensional **space surrounding**  $x$ .

- We consider this region to be a hypersphere  $V_\epsilon(x)$  centered at  $x$ , whose radius  $\epsilon$  is a **user-defined parameter**.
- Let  $N_\epsilon(x)$  denote the number of points of  $X$  lying in  $V_\epsilon(x)$ .
- Another **user-defined parameter** is the minimum number of points,  $q$ , that must be contained in  $V_\epsilon(x)$ , in order for  $x$  to be considered an **interior** point of a cluster.

# Definitions

## Definition

A point  $y$  is **directly density reachable** from a point  $x$  if

- $y \in V_\epsilon(x)$
- $N_\epsilon(x) \geq q$

## Definition

A point  $y$  is **density reachable** from a point  $x$  in  $X$  if there is a sequence of points  $x_1, x_2, \dots, x_p \in X$ , with  $x_1 = x$ ,  $x_p = y$ , such that  $x_{i+1}$  is **directly density reachable** from  $x_i$

## Definitions

It is easy to note that if  $x$  and  $y$  in the previous definition are **dense enough** (that is,  $N_\epsilon(x) \geq q$  and  $N_\epsilon(y) \geq q$ ), the density reachability property is **symmetric**.

On the other hand, symmetry is lost in the case where either of the points,  $x$  and  $y$ , has less than  $q$  points in its neighborhood.

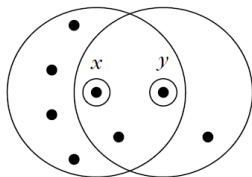
However, in this case there must be a third point  $z \in X$  such that both  $x$  and  $y$  are density reachable from  $z$ . This leads to the definition of the **density-connectivity**.

### Definition

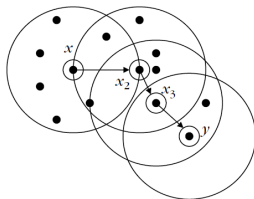
A point  $x$  is **density connected** to a point  $y \in X$  if there exists  $z \in X$  such that both  $x$  and  $y$  are **density reachable** from  $z$ .



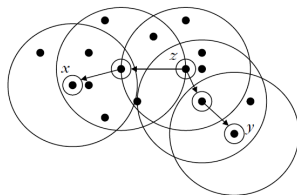
# Definitions



(a)  $y$  is **directly density reachable** from  $x$ , but not vice versa



$y$  is **density reachable** from  $x$ , but not vice versa



(b)  $x$  and  $y$  are **density connected**

# Definitions

A cluster  $C$  in the **DBSCAN framework** is defined as a nonempty subset of  $X$  satisfying the following conditions:

- **Maximality**: If  $x$  belongs to  $C$  and  $y \in X$  is **density reachable** from  $x$ , then  $y \in C$ .
- **Connectivity**: For each pair  $(x, y) \in C$ ,  $x$  and  $y$  are **density connected**.

Let  $C_1, \dots, C_m$  be the clusters in  $X$ . Then, the set of points that are not contained in any of the clusters  $C_1, \dots, C_m$  is known as **noise**.

# Definitions

A point  $x$  is defined as **core point** if it has at least  $q$  points in its neighborhood. Otherwise,  $x$  is said to be a **noncore point**.

A noncore point may be either

- A **border point** of a cluster, that is, density reachable from a core point.
- A **noise point**, that is, not density reachable from other points in  $X$ .

# Definitions

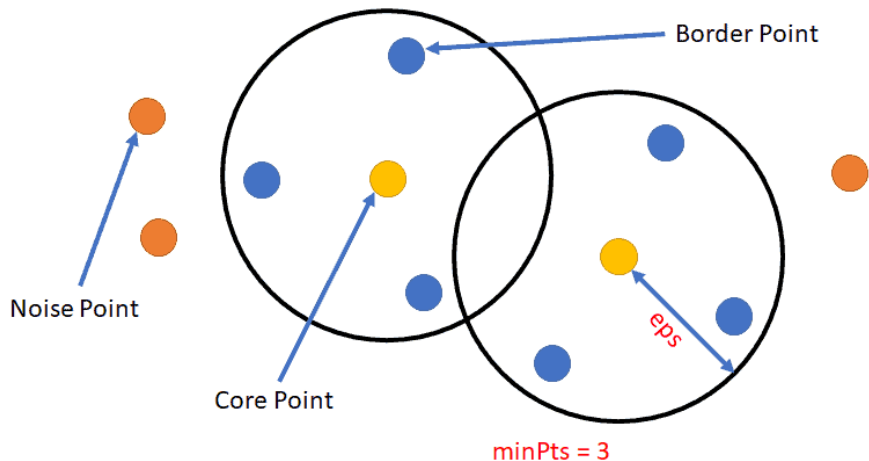


Figure: Noise, core and border point

# Proposition

The following two propositions hold true (Este, 1996).

## Proposition

If  $x$  is a core point and  $D$  is the set of points in  $X$  that are density reachable from  $x$ , then  $D$  is a **cluster**.

## Proposition

If  $C$  is a **cluster** and  $x$  a core point in  $C$ , then  $C$  equals to the set of the points  $y \in X$  that are **density reachable** from  $X$ .

These two propositions imply that a cluster is **uniquely determined** by any of its core points.

# DBSCAN Algorithm

Let  $X_{un}$  be the set of points in  $X$  that have not been considered yet, and let  $m$  denote the number of clusters.

- Set  $X_{un} = X$
- Set  $m = 0$
- While  $X_{un} \neq \emptyset$  do:
  - Arbitrarily select a  $x \in X_{un}$ .
  - If  $x$  is a **noncore point** then:
    - Mark  $x$  as noise point.
    - $X_{un} = X_{un} - \{x\}$
  - If  $x$  is a **core point** then
    - $m = m + 1$
    - Determine **all density-reachable points** in  $X$  from  $x$ .
    - Assign  $x$  and the previous points to the cluster  $C_m$ . The border points that may have been marked as noise are also assigned to  $C_m$ .
    - $X_{un} = X_{un} - C_m$
  - End if
- End while

# DBSCAN Algorithm

A delicate point of the algorithm is the following:

- Suppose that a **border point**  $y$  of a cluster  $C$  has currently been selected by the algorithm.
- This point will be marked as a **noise point**, through the first branch of the if statement of the algorithm.
- Note, however, that when, later on, a core point  $x$  of  $C$  will be considered from which  $y$  is density reachable, then  $y$  will be identified as a **density-reachable point** from  $x$  and will **be assigned** to  $C$ .
- On the other hand, if  $y$  is a **noise point** it will be marked as such and because it is not density reachable by any of the core points in  $X$  its noise label will **remain unaltered**.

## Remarks

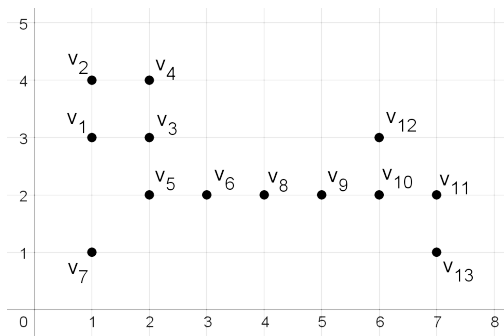
- The results of the algorithm are greatly **influenced by** the choice of  $\epsilon$  and  $q$ . Different values of the **parameters** may lead to totally different results.
- One should select these parameters so that the algorithm is able to detect the **least dense cluster**. In practice, one has to experiment with several values for  $\epsilon$  and  $q$  in order to identify the best combination for the data set at hand.
- The DBSCAN is **not appropriate** for cases where the clusters in  $X$  exhibit **significant differences** in density, and it is not well suited for **high-dimensional data**.



## Exercise

Consider the following set of patterns in  $\mathbb{R}^2$ . Apply the DBSCAN algorithm considering the Euclidean distance and the parameters  $\epsilon = \sqrt{3}$  and  $q = 5$ . Answer the following questions:

- What clusters were found by the algorithm
- What are the core and border points of each cluster?
- Identify the noise points.



# References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009